

Advanced (Habanero)

Q1. What is $f(x) + g(x)$ if $f(x) = x^2$ and $g(x) = 2x$?

- (A) $x^2 + 2x$
- (B) $x^2 - 2x$
- (C) $2x^2$
- (D) $x^2 + x$
- (E) $2x^2 + x$

Q2. If $f(x) = x - 1$ and $g(x) = x + 1$, what is $f(x) - g(x)$?

- (A) -2
- (B) 2
- (C) $x - 2$
- (D) $x + 2$
- (E) $2x$

Q3. What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$ when combined?

- (A) $x \neq 0, x \neq 1$
- (B) $x > 0, x > 1$
- (C) $x < 0, x < 1$
- (D) $x \geq 0, x \geq 1$
- (E) $x \leq 0, x \leq 1$

Q4. Given $f(x) = x^2$ and $g(x) = 2x$, what is $(f(x) \cdot g(x))'$?

- (A) $2x^2 + 4x$
- (B) $2x^2 - 4x$
- (C) $4x^2 + 2x$
- (D) $4x^2 - 2x$
- (E) $2x^2$

Q5. What is the result of $(f(x) + g(x))'$ if $f(x) = x^2$ and $g(x) = 2x$?

- (A) $2x + 2$
- (B) $2x - 2$
- (C) $4x + 2$
- (D) $4x - 2$
- (E) $2x$

Q6. If $f(x) = \frac{1}{x}$, what is $f(x) \cdot f(x)$?

- (A) $\frac{1}{x^2}$
- (B) $\frac{1}{x}$
- (C) x^2
- (D) x
- (E) 1

Q7. Given $f(x) = x^2$ and $g(x) = x + 1$, what is $f(x) - g(x)$?

- (A) $x^2 - x - 1$
- (B) $x^2 + x + 1$
- (C) $x^2 - x + 1$
- (D) $x^2 + x - 1$
- (E) $x + 1$

Q8. If $f(x) = x - 1$ and $g(x) = x + 1$, what is $f(x) \cdot g(x)$?

- (A) $x^2 - 1$
- (B) $x^2 + 1$
- (C) $x^2 - 2x + 1$
- (D) $x^2 + 2x + 1$
- (E) $2x$

Open Ended Questions (FRQ)

Q9. Given $f(x) = x^2$ and $g(x) = 2x$, determine the domain of $f(x) + g(x)$ and $f(x) - g(x)$. Show all work.

Q10. If $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$, find $(f(x) \cdot g(x))'$ and simplify. Provide all steps.

Chef's Tips

- Emphasize the importance of checking the domain of combined functions.
- Ensure students understand how to apply the product rule and quotient rule for differentiation.
- Remind students to simplify expressions fully after performing operations.